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Paleomagnetism and Age of the Archean Usushwana Complex, Southern Africa

PAUL W. LAYER[1], ALFRED KRÖNER[2], MICHAEL MCWILLIAMS[1], AND ALEXANDER BURGHELE[3]

Paleomagnetic and $^{40}Ar/^{39}Ar$ age data are reported from the Usushwana Complex, a layered gabbro and pyroxenite intrusive exposed in Swaziland and eastern South Africa (26.5°S, 31.0°E). The main gabbro body and an associated granophyre return a mean Rb-Sr whole-rock and Sm-Nd mineral age of 2875±40 Ma. Alternating field and thermal analysis of specimens from 18 sites (70 samples) yields a mean direction $D = 304°$, $I = 53°$, ($k = 32$, $\alpha_{95} = 6°$). This direction is significantly different from that obtained from the ~ 2100 Ma Bushveld Complex. Rare dikes of probable Bushveld age cut the Usushwana Complex, and a low coercivity overprint seen in specimens from a few sites has a Bushveld direction and almost certainly reflects this event. Thermal overprinting is recorded by $^{40}Ar/^{39}Ar$ hornblende and pyroxene ages between 2094±54 Ma and 2386±58 Ma that we interpret to have occurred at low temperature but over a significant period of time. Isothermal remanent magnetization acquisition experiments and laboratory unblocking temperatures suggest that the primary remanence carrier is single domain magnetite. The paleomagnetic pole position calculated from the Usushwana Complex (9°N, 347°E) is significantly different from other late Archean poles. Although the Usushwana pole is close to the pole from the Palabora Igneous Complex of South Africa (1900 Ma), there is no isotopic evidence indicating a thermal event of that age in the Usushwana Complex.

INTRODUCTION AND REGIONAL SETTING

Paleomagnetic study of reliably dated rocks from stable cratonic regions is fundamental to an understanding of the behavior of the Archean geomagnetic field and of ancient plate motions. The apparent polar wander path for the Kaapvaal and Zimbabwe (Rhodesian) cratons of southern Africa extends to approximately 2600 Ma with a paleomagnetic pole from the Modipe gabbro of Botswana [*Evans and McElhinny*, 1966]. (Isotopic ages quoted in this paper are based on the new decay constants of *Steiger and Jäger* [1977]; isotopic ages using older constants were recalculated where necessary; all quoted errors are 2σ unless otherwise noted.) This study of the Usushwana Complex in Swaziland is part of more comprehensive studies by several research groups of dated rocks from the eastern Kaapvaal Craton in which we hope to extend the apparent polar wander path for this part of Africa considerably past 3000 Ma [*Hale and Dunlop*, 1984; *Layer et al.*, 1983].

The Kaapvaal Craton is the Archean basement of southeastern Africa (Figure 1). Together with the Zimbabwe Craton and the intervening Limpopo Mobile Belt, the Kaapvaal Craton probably existed as a single structural unit for much of Archean time [*Barton and Key*, 1981]. It is best exposed in its eastern part as a granitoid-greenstone-gneiss terrane exposed in an approximately 100-km-wide north-south trending segment bounded by gently dipping early Proterozoic (Transvaal) and Paleozoic to Mesozoic (Karoo) sedimentary and volcanic strata (Figure 2).

The oldest known rocks in this area are the bimodal suites of the Ancient Gneiss Complex (AGC) of Swaziland and the Barberton Greenstone Belt (BGB) of South Africa and Swaziland [*Tankard et al.*, 1982]. The AGC has Rb-Sr and Sm-Nd whole rock isochron ages of 3555±111 Ma and 3450±40 Ma, respectively [*Barton et al.*, 1980; *Carlson et al.*, 1983], while komatiitic to felsic lavas of the BGB have been dated by the Sm-Nd whole rock method at 3540±30 Ma and 3560±230 Ma [*Hamilton et al.*, 1979; *Jahn et al.*, 1982]. The relationship between the AGC and the BGB is uncertain; however, both were intruded by granitoid plutons generated during magmatic episodes between 3500 and 3200 Ma. The widespread Mpuluzi Pluton (formally the Lochiel Pluton), a coarse-grained, homogeneous, potassium-rich granite [*Anhaeusser and Robb*, 1981], lies between the AGC and BGB and was emplaced after these magmatic episodes at 3028 ± 14 Ma [*Barton*, 1983]. Deposition of the Pongola Supergroup (2940±22 Ma [*Hegner et al.*, 1984]) followed the intrusion of the Mpuluzi Pluton. The Pongola is composed of volcanics and shallow water shelf deposits and is subdivided into the lower Insuzi and overlying Mozaan groups. The Usushwana Complex intrudes the older greenstones and crystalline rocks as well as Pongola Supergroup rocks and is in turn cut by the Ngwempisi Pluton (2802±56 Ma) and the Sicunusa Pluton (2608 ± 123 Ma) [*Barton*, 1983]. Following the emplacement of these post-Usushwana granitic plutons, the complex was cut by minor diabase dikes of probable Bushveld age at approximately 2100 Ma. No major tectonic activity seems to have affected the complex since this time.

GEOLOGIC SUMMARY

The name "Usushwana Complex" was first applied to a suite of gabbroic rocks between the Usushwana and Usutu rivers in Swaziland [*Hunter*, 1970]. Later studies extended the Complex and included similar rocks in South Africa which intrude the Insuzi Group. The following summary has been adapted from *Hunter* [1970].

The Usushwana Complex is a crescent-shaped body com-

[1]Department of Geophysics, Stanford University, Stanford, California.
[2]Institut für Geowissenschaften, Johannes Gutenberg Universität, Mainz, Federal Republic of Germany.
[3]Max-Planck-Institut für Chemie, Mainz, Federal Republic of Germany.

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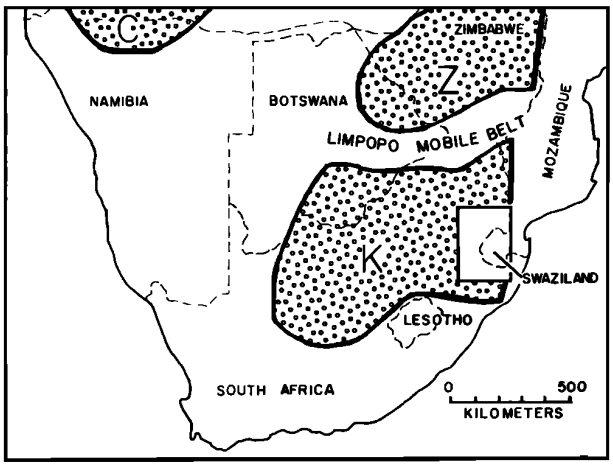


Fig. 1. Sketch Map of southern Africa showing the major Archean cratons and the Limpopo Mobile Belt. C, Congo Craton; K, Kaapvaal Craton; Z, Zimbabwe Craton. The box at the eastern edge of the Kaapvaal Craton refers to the area shown in Figure 2.

prised of two dikelike arms about 70 km apart (Figure 2). The southern arm extends into South Africa, where it intruded the Insuzi Group as a gabbro/granophyre sill. The northern arm, west of Mbabane, Swaziland, is the best exposed and mapped part of the complex and was thus the focus of our study (Figure 3). Here the complex intrudes the Mpuluzi Pluton, has a maximum width of about 6.5 km, and trends 40 km northwest from the Usutu River to the Transvaal border. In the northwest it cuts greenstones of the Onverwacht Group in the BGB, while in the southeast it is intruded by the Ngwempisi Pluton. In the south central part of this arm, andesitic lavas and quartzites of the Insuzi Group form a roof pendant in the complex. The main body of the complex is primarily composed of a medium-grained quartz gabbro with some evidence of layering. At the western edge of the complex is a coarse-grained pyroxenite composed of olivine, hypersthene, augite, some interstitial plagioclase, and flecks of red-brown biotite. Toward the center of the complex, the pyroxenite becomes more fine-grained with numerous rounded grains of olivine, hypersthene and clinopyroxene [*Winter*, 1965].

Gravity studies of the complex in the central portion of the northern arm suggest that it is a steep-sided dike, similar to the Great Dyke of Zimbabwe [*Hunter*, 1970]. Contacts between the country rock and the gabbro are not well exposed, so the exact nature of the margins of the complex are not known. Eight distinctive cumulate layers have been identified in surface exposures in parts of the complex [*Winter*, 1965; *Hunter*, 1970]. Near Mlambanyati, borehole logs show that these individual cumulate layers dip symmetrically at about 45° toward the center of the complex, while the layering is less symmetric to the north. Layering was not recognized in the areas sampled for paleomagnetic study. *Hunter* [1970] proposed a three-stage model for the development of the complex in which the pyroxenite and some of the hypersthene gabbros were deposited as horizontally layered intrusions through various feeders. Quartz gabbro was next intruded, reheating and tilting the earlier intrusions. This flooding of the earlier layers caused the asymmetric layering in the north. Finally, the dike subsided in a fault-bounded graben with little or no rotation.

A significant problem in the paleomagnetic study of intrusive bodies such as the Usushwana Complex is estimating the effect of postintrusive tectonic movement. The Pongola Supergroup sediments which outcrop near the Usushwana Complex could be used as stratigraphic control, but in Swaziland, much of the Pongola Supergroup has been invaded by younger granitic plutons which have locally deformed the strata into a series of folds with dips as great as 60° [*Hunter*, 1963]. In South Africa, south of the southwest arm of the complex, the Pongola sediments are less deformed and have dips of 5°–15°, about north to northeast fold axes [*Matthews*, 1967]. This and the absence of exposed lower crustal rocks suggest that the eastern Kaapvaal Craton has not been tilted significantly since 2800 Ma. For the Usushwana Complex the symmetry of the layering and its fault-bounded vertical sides as indicated by gravity data are the best arguments that we can present to support the contention that the complex has not been tilted considerably in relation to neighboring rock units.

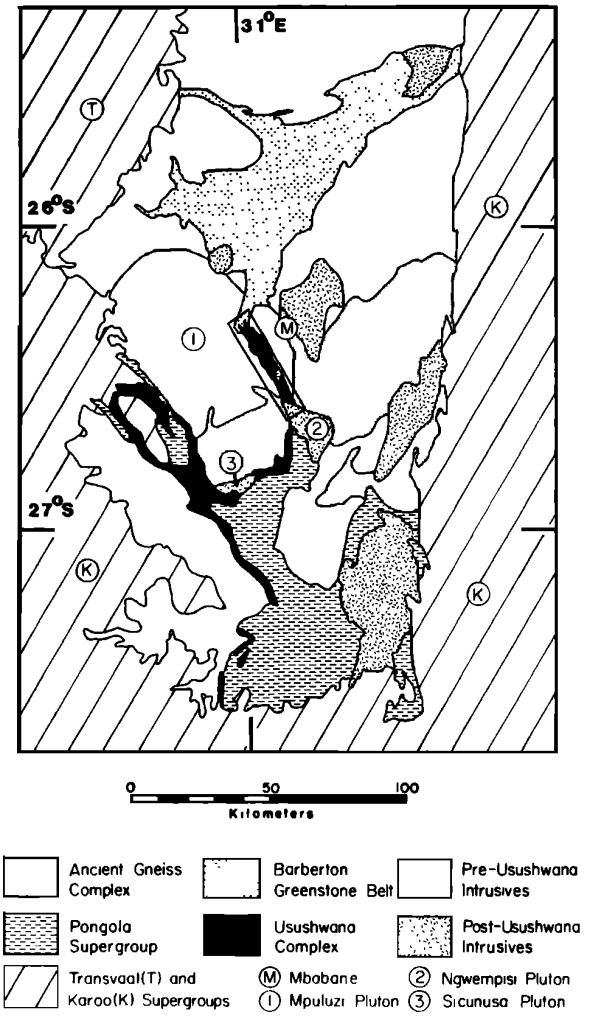


Fig. 2. Generalized geologic map showing Precambrian exposures of the Kaapvaal Craton in Swaziland and eastern South Africa. The Usushwana Comples is shown in the lower half of the figure. The box around the northwestern arm of the Complex refers to the area shown in Figure 3.

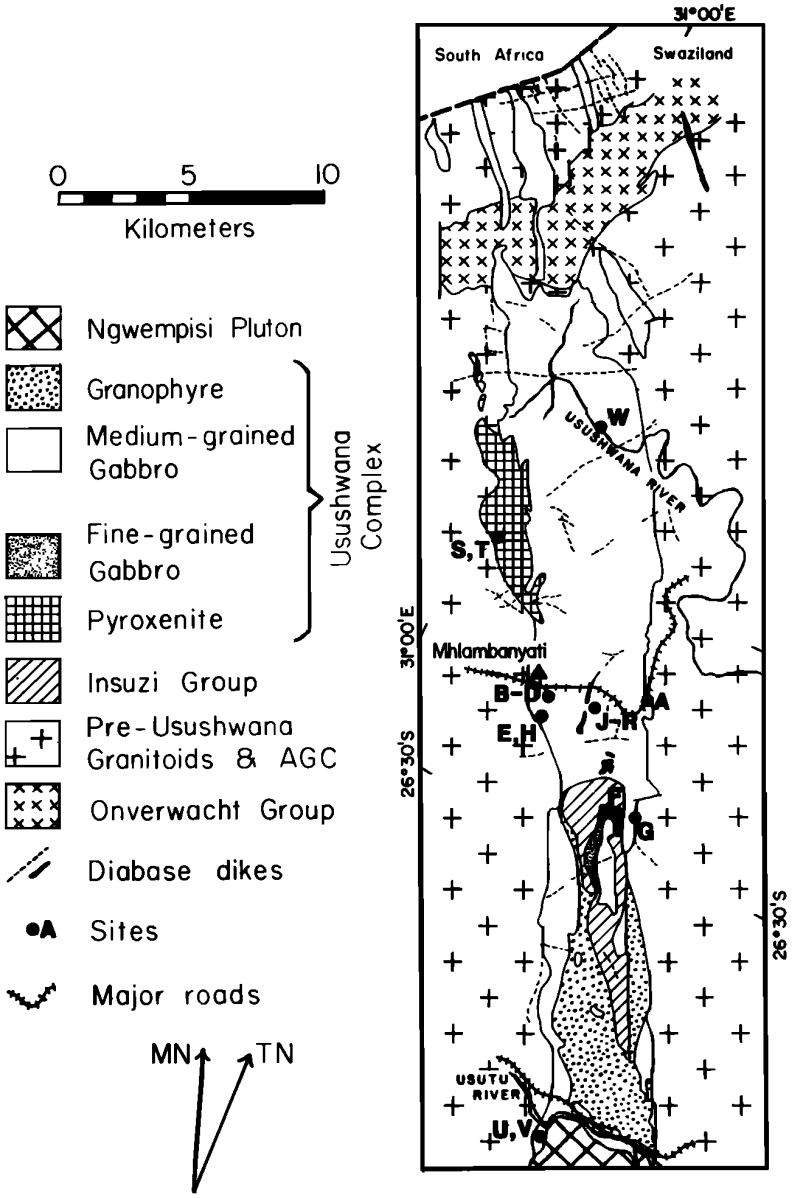


Fig. 3. Generalized geologic map of the Usushwana Complex in northeastern Swaziland, adapted from Hunter [1970] and shows site locations (solid dots). The Onverwacht Group is a unit in the BGB; the Insuzi Group is the lowest unit of the Pongola Supergroup.

Isotopic Ages

Two independent Rb-Sr whole rock isochron ages were obtained from the medium-grained gabbro near Mlambanyati (where most paleomagnetic samples were collected) and from a granophyre in the south [*Davies et al.*, 1970; *A. Tegtmeyer, unpublished data*, 1982] The initial $^{87}Sr/^{86}Sr$ ratios (I_{Sr}) for both isochrons are identical within experimental error, and the ages obtained are 2882±52 Ma and 2870±38 Ma, respectively (Table 1). An internal Sm-Nd isochron for a pyroxenite with fresh clinopyroxene, orthopyroxene, and plagioclase yielded an age of 2871±30 Ma [*Hegner et al.*, 1984] (Table 1). In view of the excellent agreement of the Sm-Nd age with the Rb-Sr ages, we consider a mean age of 2875±40 Ma to reflect the crystallization age of the rocks. This good agreement of ages and the fact that a regional dating survey by *Barton* [1983] indicate that no large-scale regional metamorphism is seen in the eastern Kaapvaal Craton since about 3000 Ma support the conclusion that the age of magnetization is not significantly younger than 2875±40 Ma.

To test this conclusion, hornblende and pyroxene concentrates from sites L and P were dated by the $^{40}Ar/^{39}Ar$ step-heating technique at the Max-Plank-Institut für Chemie following techniques detailed by *Burghele* [1980]. Hand-picked mineral separates (0.2–1.0 mm in size) were irradiated with fast neutron flux, totaling a neutron dose of 3×10^{19} cm^{-2}. The $^{40}Ar/^{39}Ar$ ages are normalized to the Bern Standard 4M, which was irradiated together with the samples and has an age of 18.555±0.395 Ma and $^{40}Ar/^{39}Ar = 4.4411 \times 10^{-2}$. The measurements were performed at temperature steps starting at 300°C up to sample fusion. Correction for

Table 1. Rb-Sr and Sm-Nd Radiometric Ages

Method	Rock Type	Age $\pm 2\sigma$	Reference
Rb-Sr whole rock	gabbro and granophyre	2882 ± 52 $I_{Sr} = 0.7023 \pm 0.0013$	Tegtmeyer [1982] (unpublished data)
Rb-Sr whole rock	gabbro, granophyre and pegmatite	2870 ± 38 $I_{Sr} = 0.7009 \pm 0.0136$	Davies et al. [1970] (recalculated)
Sm-Nd whole rock and mineral separates	pyroxenite	2871 ± 30 $I_{Nd} = 0.50876 \pm 0.00003$	Hegner et al. [1984]

reactor-produced ^{40}Ar and the Ca correction were performed. Results are presented in Table 2 and plotted in Figure 4.

All samples exhibit severe argon loss from the mineral systems since the integrated ages are distinctly lower than the Sm-Nd and Rb-Sr ages. The plateaus at relatively low to intermediate degassing temperatures (700° – 1150°C) shown by pyroxene sample L at 2293±58 Ma (Figure 4*b*) and P at 2230±58 Ma (Figure 4*c*) are significant features in the Ar release spectra of this mineral. In addition, both pyroxene samples and the hornblende sample exhibit anomalously high gas release at low temperatures and variable $^{37}Ar/^{40}Ar$ ratios (indicative of variable Ca/K ratios). These features indicate that alteration of the minerals occurred, possibly

Table 2. The $^{40}Ar/^{39}Ar$ Step-Heating Mineral Ages

Temperature, °C	^{40}Ar (radiogenic), $\times 10^{-8}$ cm^3/g STP	^{39}Ar, $\times 10^{-8}$ cm^3/g STP	^{37}Ar, $\times 10^{-8}$ cm^3/g STP	Age, Ma $\pm 1\sigma$
		Sample L: Hornblende		
300	107.3520±2.71%	7.1762±1.09%	9.4119±0.89%	2706±74
450	32.7856±6.09%	7.9345±0.45%	6.0068±1.52%	1216±67
585	145.3920±1.06%	43.2486±0.46%	26.5656±0.52%	1032±36
708	526.7570±0.70%	60.0035±0.67%	29.6658±0.64%	2008±54
820	492.6880±0.76%	56.1907±0.69%	48.2775±0.58%	2007±55
922	978.5110±0.56%	115.2660±0.57%	242.3480±0.48%	1968±53
1015	1063.4500±0.54%	118.1570±0.58%	1640.1800±0.58%	2038±54
1100	1324.6000±0.50%	124.6370±0.48%	2191.9400±0.50%	2247±57
1178	522.2750±0.51%	49.2572±0.46%	711.9920±0.43%	2244±57
1249	184.7080±0.73%	16.9301±0.45%	321.6060±0.48%	2280±58
1314	518.2190±0.54%	34.5406±0.47%	1485.6500±0.45%	2710±62
1374	93.8809±1.29%	6.2102±0.54%	313.6360±0.45%	2721±65
1429	84.6042±1.72%	5.9477±0.66%	343.4900±0.46%	2636±66
1479	9.8267±9.73%	0.7044±1.44%	37.4928±0.48%	2609±148
Total	6083.0600±0.21%	646.2040±0.20%	7408.2700±0.22%	2094±54
		Sample L: Pyroxene		
300	66.1249±1.74%	4.5808±0.51%	11.4680±0.59%	2657±66
585	153.5600±0.91%	27.0287±0.40%	22.4772±0.50%	1520±46
708	308.5950±0.64%	34.1639±0.44%	21.5599±0.71%	2043±55
820	228.6200±0.55%	23.0005±0.37%	31.6874±0.49%	2161±56
922	342.4840±0.47%	35.5562±0.38%	190.2510±0.41%	2122±55
1015	829.3950±0.53%	75.2649±0.53%	1203.4500±0.45%	2293±58
1100	187.4090±0.69%	14.8598±0.39%	266.3670±0.44%	2472±60
1178	115.1370±0.76%	9.6819±0.43%	208.3580±0.39%	2393±59
1249	189.4940±0.63%	11.6181±0.47%	634.4800±0.39%	2829±64
1314	463.9170±1.16%	12.6628±0.57%	8013.0300±0.41%	4067±74
1374	44.3202±2.08%	0.8575±0.80%	938.2480±0.39%	4632±82
Total	2929.0600±0.27%	249.2550±0.19%	11541.3000±0.29%	2377±58
		Sample P: Pyroxene		
300	34.5080±9.59%	3.3399±0.52%	13.7992±3.71%	2211±146
585	127.0630±2.21%	23.8964±0.40%	34.0674±1.09%	1453±49
820	448.9200±1.17%	52.3069±0.39%	81.8843±0.71%	1981±55
1015	1274.4000±0.62%	121.1210±0.41%	1503.6100±0.43%	2234±57
1178	354.3120±1.33%	33.7838±0.50%	572.8270±0.37%	2230±59
1314	358.4330±1.32%	24.9282±0.39%	1283.1400±0.38%	2651±64
1429	659.4840±1.15%	16.0536±0.48%	7623.6500±0.38%	4253±74
Total	3257.1100±0.45%	275.4300±0.21%	11113.0000±0.27%	2386±58

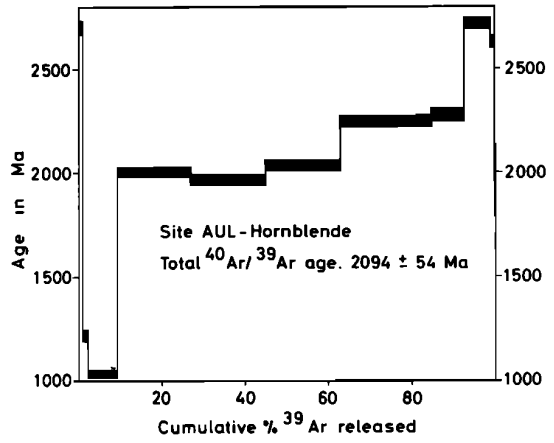


Fig. 4*a*. Age spectrum of hornblende concentrate from Usushwana sample *L*. Width of black bars represents 1σ errors; for data see Table 2.

due to a thermal event, causing mica lamellae to develop in the hornblendes and pyroxenes. These lamellae can be seen in thin section. The reheating may be related to the early phases of Bushveld magmatism, and we shall demonstrate below that some of our samples may have been magnetically reset by the same event.

PALEOMAGNETIC STUDY

Samples oriented with sun and magnetic compass were collected at 22 sites in the northern arm of the Usushwana Complex (Figure 3). All sites were located at natural outcrops, except site A which was located at a roadcut. All but three sites (F,S,T) were from the medium-grained quartz gabbro. Site F was from a fine-grained gabbro, and sites S and T were from the fine-grained pyroxenite.

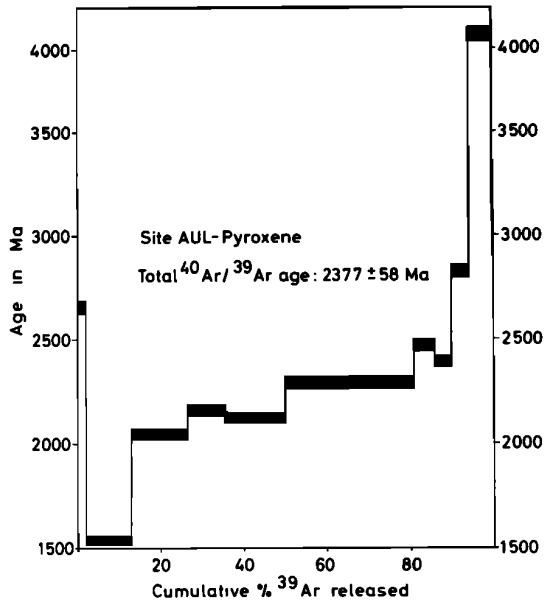


Fig. 4*b*. Age spectrum of pyroxene concentrate from Usushwana sample *L*. Width of black bars represents 1σ errors; for data see Table 2.

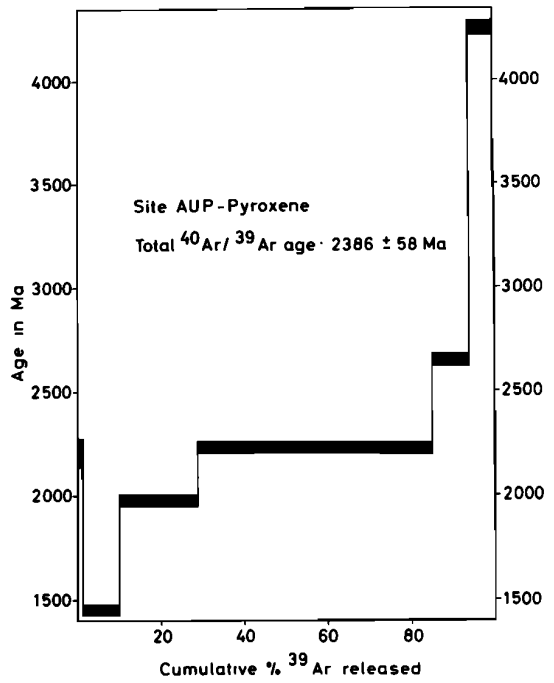


Fig. 4*c*. Age spectrum of pyroxene concentrate from Usushwana sample *P*. Width of black bars represents 1σ errors; for data see Table 2.

Magnetic measurements were made on a two-axis superconducting magnetometer. Alternating field (AF) demagnetization was done inside the magnetometer using a two-coil system. Thermal demagnetization was performed in air and the samples were cooled in a shielded chamber with static fields less than 5 nT. Thermally demagnetized samples were subjected to a 5 mT peak alternating field after cooling to remove any viscous magnetization acquired between the cooling chamber and magnetometer.

One specimen cut from each sample was subjected to stepwise AF demagnetization in peak fields up to 80 mT, usually in 10-mT steps. Specimens from a few sites showed particularly high coercive force spectra; these were further stepwise AF demagnetized to 230 mT using a four-axis tumbler demagnetizing system. Specimens from three sites (G,H,V) showed erratic directional behavior during AF demagnetization and are not considered further.

Demagnetization of representative specimens from the remaining 19 sites suggests that three characteristic responses of the magnetization vector to progressive AF demagnetization can be described. One group of specimens (sites A, C, W) exhibits a single component of magnetization with a coercive force ranging from 10 to 50 mT (Figure 5*a*). For these specimens, a linear fit to the demagnetization vector was determined using a least squares technique similar to *Kirschvink* [1981]. The second group of specimens (sites D, E, P) shows a coercive force exceeding 80 mT (Figure 5*b*). Progressive thermal demagnetization of these specimens isolated a magnetization direction with unblocking temperatures of about 580°C (Figure 5*c*). During thermal treatment, magnetization directions did not change significantly, implying that a single magnetization is present. Least squares linear techniques were used to obtain the magnetization direction for these specimens.

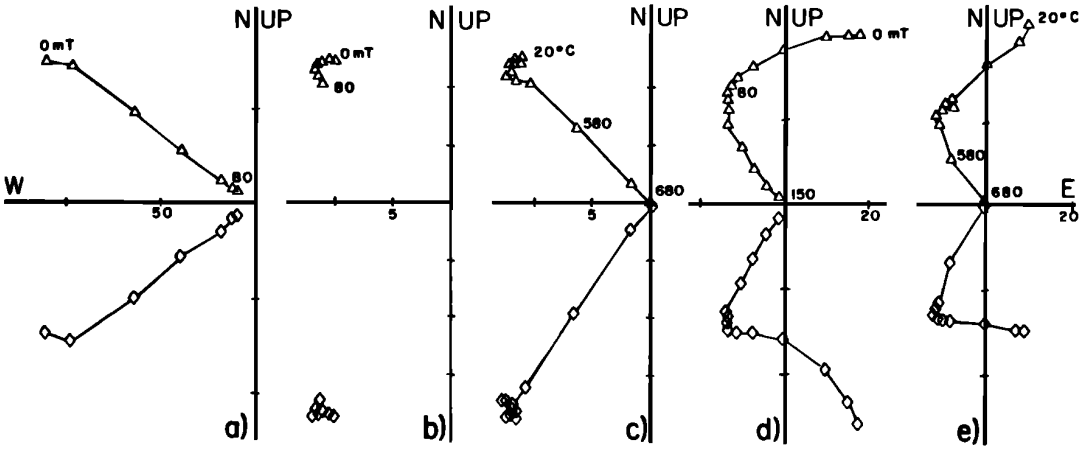


Figure 5. (*a*) A typical orthogonal vector plot for the first type of magnetization (site A) showing a single magnetization removed during AF demagnetization. (*b* and *c*) Typical orthogonal vector plots of the second type of magnetization observed (site P) showing (*b*) AF and (*c*) thermal demagnetization experiments. (*d* and *e*) Typical orthogonal vector plots of the third type of magnetization observed (site R) showing (*d*) AF and (*e*) thermal demagnetization experiments.

Specimens from the remaining 13 sites show coercive forces ranging from 30 to 80 mT. Natural remanent magnetization (NRM) directions for specimens from these sites were scattered, but during AF demagnetization their vectors path described great circle segments when plotted on a stereographic projection. At peak fields above 100 mT, their directions stablized and intensity decreased, implying that only one magnetic component remained (Figure 5*d*). Thermally demagnetized specimens exhibited similar behavior to that seen in specimens from the same sample during AF demagnetization (Figure 5*e*).

A lower coercivity magnetization with shallow east to northeast declination was observed in specimens from several sites (Figures 5*d* and 5*e*). This component was removed below 60 mT and had a maximum unblocking temperature of 490°C. This component is similar in direction to a low coercivity component reported by *Hale and Dunlop* [1984] for the Komati Formation and is also seen as the primary direction in preliminary results from the Ngwempisi and Mbabane plutons [*Layer*, 1986]. It may represent overprinting due to igneous activity between 2600 and 2800 Ma.

There is also a very low coercivity component (< 20 mT) that is seen in some specimens (Figure 5*d*). Because of its similar direction, this north to northeast steep component is interpreted to be an overprint of the age of the Bushveld Complex at approximately 2100 Ma [*Gough and van Niekerk*, 1959]. A Bushveld overprint direction might be expected in the study area, as there are several small dikes of presumed Bushveld age that could have caused local reheating of the Usushwana Complex. The presence of these secondary components that range in age from 2100 to 2800 Ma may reflect partial reheating of the Usushwana Complex that may be partly responsible for the resetting of the argon system described above. It does not appear that the reheating was great enough to fully reset the magnetic system, however. Based on the agreement of the Rb-Sr and Sm-Nd dates, we feel that their age more accurately reflects the age of the magnetization, while the argon ages reflect the age of the lower-coercivity components.

In addition to the Bushveld component there are some random low-coercivity components. We attribute these to probable lightning strikes, as the complex forms high topographic features in the sampling area. The random components were generally associated with higher NRM intensities. For example, site W showed a wide range of NRM intensities from 5×10^{-2} A/m to 3.0 A/m. The two specimens with the highest intensities also had stable, but anomalous, remanence directions (with respect to the directions from the other eight samples from the site and the overall mean direction) which also may be due to lightning.

The site mean directions obtained by AF demagnetization are summarized in Table 3 and Figure 6. Thermal demagnetization experiments served to confirm the direction isolated by AF methods. Site U is located far to the south of the other sites, close to the contact with the younger Ngwempisi Pluton, and the site mean direction from specimens at this site has a significant steeper inclination than those obtained for site means farther to the north. The site U mean direction is included in the calculation of the overall mean direction (Table 3), but we do not know if the Ngwempisi Pluton is responsible for the anomalous direction. After AF demagnetization, the mean direction from site F is significantly different from that of the other sites. Thermal demagnetization of specimens from this site exhibited unstable behavior which was inconsistent with the results obtained from AF experiments. For this reason, results from site F are not included in the calculation of the overall mean direction. The overall mean direction based on the remaining 18 sites ($D = 304°$, $I = 53°$) differs by more than 100° from the earth's present field direction ($D = 344°$, $I = -64°$) or the expected direction for an axial geocentric dipole at the sampling site ($D = 0°$, $I = -45°$) (Figure 6). The k value of 32 suggests that paleosecular variation (PSV) might have been averaged, but this suggestion is based upon Phanerozoic observations of PSV, and we have no idea of what an expected dispersion value might be for the Archean.

MAGNETIC MINERALOGY

Thermal demagnetization experiments reveal unblocking temperatures indicative of titanium-poor magnetite (Figure 7*a*). Isothermal remanent magnetization (IRM) acquisition

Table 3. Paleomagnetic Data

Site	D	I	N	κ	α_{95}	λ	ϕ
A[a]	304.0	26.5	3(3)	696.6	4.7	22.2	330.7
B[b,c]	306.0	45.0	3(3)	-	-	15.7	342.3
C[a]	304.9	50.7	3(3)	622.4	4.9	11.8	345.4
D[a]	304.2	54.3	3(3)	1109.6	3.7	9.1	343.4
E[a]	298.8	52.6	3(3)	182.1	9.2	6.7	343.4
F[a,c,d]	191.5	72.6	4(4)	97.4	9.4	-57.5	19.6
J	312.3	57.3	4(4)	261.8	5.7	11.6	354.4
K	293.2	41.1	3(4)	65.1	15.4	8.3	332.6
L	304.5	48.4	4(4)	5697.1	1.2	12.9	343.6
M	313.0	57.0	4(4)	61.0	11.9	12.2	354.6
N	320.4	56.3	4(4)	42.1	14.3	16.5	358.9
O	305.6	54.6	4(4)	101.2	9.2	9.7	348.6
P	311.8	48.4	4(4)	416.4	4.5	17.5	348.1
Q	329.0	53.4	4(4)	125.7	8.2	22.8	3.4
R	331.4	44.9	4(4)	561.3	3.9	30.3	1.3
S[e]	275.4	58.0	4(4)	51.9	12.9	-12.3	338.3
T[e]	279.6	61.0	4(4)	132.2	8.0	-10.8	342.8
U	273.9	80.4	4(4)	108.0	8.9	-23.9	10.6
W	283.1	42.2	8(10)	6.0	24.8	0.0	328.5
Average[d]	304.3	52.7	18	32.2	6.2		
Average[d]			18	22.0	7.6	9.2	347.0

D, site mean declination; I, site mean inclination; N, number of specimens used to calculate site mean direction. The number in parentheses is the number of independently oriented specimens collected from a site or obtained from a single hand specimen (for sites A—F). κ, precision parameter; α_{95}, radius of the circle of 95% confidence; λ and ϕ, latitude and longitude of virtual geomagnetic poles (north and east positive).

[a]Samples A, B, C, D, E, F, and G are oriented hand samples.

[b]Site direction determined using converging remagnetization circles [Halls, 1976].

[c]Fine grained gabbro.

[d]Average includes all sites except Site F.

[e]Pyroxenite.

experiments were performed on samples from the medium-grained gabbro and the fine-grained pyroxenite (Figure 7*b*). The samples used in the IRM experiments were first demagnetized in a peak AF field of 80 mT. IRM of the medium-grained samples became saturated at around 300 mT, while the fine-grained samples saturated at about 100 mT. These results are similar to those reported for the Modipe Gabbro of Botswana by *Evans et al.* [1968], who found that the carriers of the highly stable remanence were elongate single domain magnetite particles with semi-major axis lengths of 5–10 μm situated within pyroxene grains.

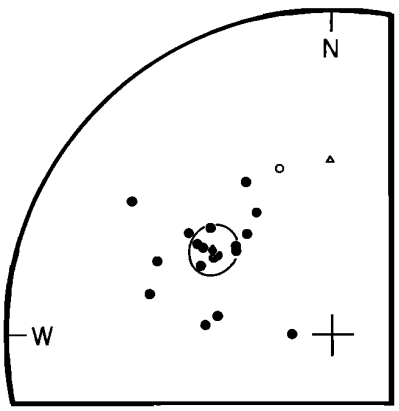


Fig. 6. Equal-area stereographic projection of site mean directions from 18 sites that were used in final analysis; open (closed) symbols represent negative (positive) inclinations. The mean direction for all sites is shown by the solid diamond; the open circle corresponds to the present field direction; the open triangle corresponds to the axial geocentric dipole direction; the oval represents the α_{95} associated with the overall mean direction.

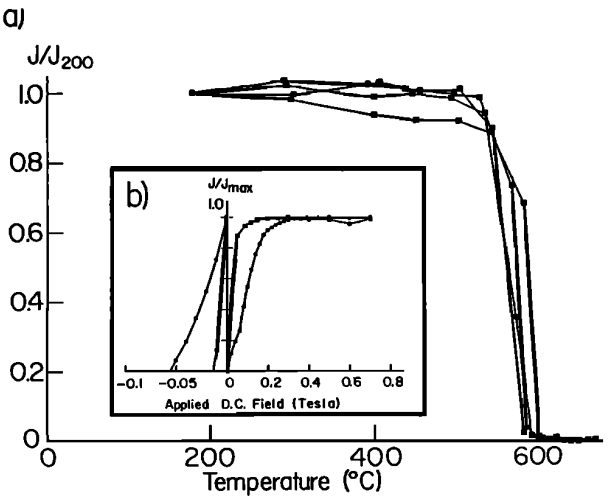


Fig. 7. (*a*) Plot of normalized intensity (J/J_{200}) versus temperature for representative samples of the medium-grained gabbro. (*b*) IRM acquisition experiments for a fine-grained sample (squares) and for a medium-grained sample (circles). The curves are normalized to the maximum intensity observed for each sample. The reverse field required to destroy the saturation magnetization of each sample is shown to the left.

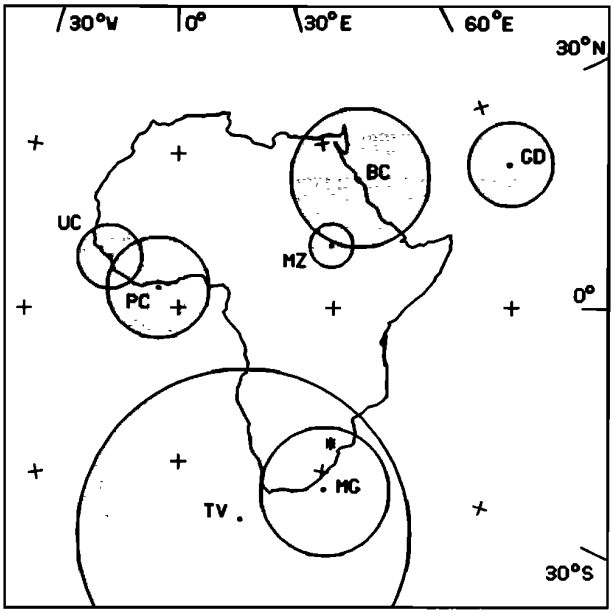


Fig. 8. Pole positions and A_{95} circles for the Usushwana Complex (UC, 2850±50 Ma), the Modipe Gabbro (MG, 2575±470 Ma), the Great Dyke (GD, 2460±16 Ma), the Transvaal Supergroup (TV, 2250 Ma), the Bushveld Complex (BC) and the Main Zone of the Busheveld Complex (MZ, 2095±24 Ma), and the Palabora Igneous Complex (PC, 1900 Ma). The location of the Usushwana Complex is shown by the asterisk.

Thin sections were examined from sites L and P in the gabbro and site S in the pyroxenite. All three rocks showed minor signs of postcrystallization alteration. Microprobe scanning of large (1–3 μm) opaque grains identified mainly ilmenite with lamellae of sphene and chromite. No large magnetite grains were seen in the gabbro, implying that the magnetic carriers are smaller than about 1 μm, and perhaps occur as needles in the pyroxenes, as *Evans et al.* [1968] reported. Some of the large opaque grains seen in the pyroxenite showed magnetite zones which may be the cause of the high-intensity NRM component. The presence of visible (presumably multidomain) magnetite in the finer-grained samples may explain why the IRM of those samples became saturated in a lower field than the medium-grained ones. This is contrary to what might be expected if the size of the magnetite grains was proportional to the size of other nonmagnetic grains and implies that these visible grains are not carrying the high-coercivity remanence.

DISCUSSION

A 2875±40 Ma paleomagnetic pole for the Kaapvaal Craton can be calculated from the mean direction obtained for the Usushwana Complex (UC), assuming no postcooling tilt, an axial geocentric dipole field model and that the Rb-Sr and Sm-Nd ages do not predate significantly the magnetic blocking temperature. This pole (9°N, 347°E, A_{95} = 8°) is plotted together with five previously reported poles for the Kaapvaal Craton in Figure 8. The pole for the Modipe Gabbro of Botswana (MG) is from *Evans and McElhinny* [1966] and has a poorly constrained age. *McElhinny* [1966] reported a Rb-Sr whole rock errorchron age of 2575±470 Ma and mean pyroxene $^{40}Ar/^{39}Ar$ age of 2654±100 Ma for the Modipe Gabbro. The pole for the Great Dyke of Zimbabwe (GD) is from *Jones et al.* [1975] with an emplacement age of 2460±16 Ma [*Hamilton*, 1977]. The pole from gabbros of the Bushveld Complex in South Africa (BC) is from *Gough and van Niekerk* [1959]. An additional pole from the Main Zone of the Bushveld Complex (MZ) is from *Hattingh* [1983]. These gabbros have a Rb-Sr age of 2095±24 Ma and an ^{40}Ar-^{39}Ar age of 2098±12 Ma [*Tankard et al.*, 1982]. The pole from the Transvaal sequence (TV) is from *Briden* [1976] and has an age of approximately 2250 Ma.

The pole for the Palabora Igneous Complex of South Africa (PC) is from *Morgan and Briden* [1981] and has an associated age of ~ 1900 Ma. Transvaal to Bushveld age dolerite dikes cut the Usushwana Complex, but the pole that we obtained is significantly different from those of the Transvaal sequence and Bushveld Complex. If the poles originally coincided, a rotation of the Usushwana Complex of about 55° counterclockwise about a vertical axis or 55° about a northwest-southeast horizontal axis since 2000 Ma is necessary to move the pole to its present location. There is no evidence of either type of block rotation anywhere in central Swaziland, and thus it does not appear that the Bushveld event had any significant thermal effect on the samples of the Usushwana Complex.

It is possible that some alteration occurred at 2000 Ma. This can be seen in the disturbed Ar spectra and in overprinting of the magnetic system. However, the undisturbed Rb-Sr age, the presence of late Archean overprints due to the intrusion of granitic plutons elsewhere and the difference between the UC pole and the BC pole lead to the conclusion that the magnetization seen in the Usushwana Complex predates the intrusion of the Bushveld Complex.

The pole that we obtained from the Usushwana Complex is similar to that of the Palabora Complex and also lies near the apparent polar wander path for Africa for 500 Ma (Hook intrusives) and 1050 Ma (Kisii lavas) [*McWilliams*, 1981]. The available isotopic data indicate that no major thermal overprints of these ages have affected the Usushwana Complex, and we suggest therefore that the Usushwana Complex pole may be primary. If the magnetization is indeed primary, the difference in direction between the Usushwana Complex pole and the other poles suggests that there was significant motion of the Kaapvaal Craton in the Archean with respect to the geomagnetic reference frame. The minimum rms velocity inferred from these pole positions is, in fact, not significantly different from that obtained for North America for the same time [*Kröner et al.*, 1982].

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A. Burghele, Max-Plank-Institut für Chemie, D-6500 Mainz, Federal Republic of Germany.

A. Kröner, Institut für Geowissenschaften, Johannes Gutenberg Universität, D-6500 Mainz, Federal Republic of Germany.

P. W. Layer and M. McWilliams, Department of Geophysics, Stanford University, Stanford, CA 94305.

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